Nipuni Hansika

Karumpulli Arachchige Nipuni Hansika

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Profile

Human–Computer Interaction researcher with research experience in creativity support tools, embodied and multimodal interaction, AR/VR, and user-centered design. Co-author of a full paper published at ACM CHI 2026 on motor-mediated creativity and digital expression. Experienced in designing and developing interactive research prototypes and conducting empirical user studies using quantitative and qualitative methods. Currently a Lecturer at the University of Ruhuna, teaching Machine Learning with hands-on laboratory sessions and supervising undergraduate projects. Passionate about bridging technology, creativity, and human-centered design through research and teaching.

Research Interests: Human–Computer Interaction (HCI), Human–AI Interaction, Creativity Support Tools and Computational Creativity, AI-mediated Exploration and Learning, Human Agency and Human–Centered AI, Embodied and Multimodal Interaction, Extended Reality (AR/VR) and Immersive Interfaces, Accessible and Inclusive Computing.

Education

SINGAPORE MANAGEMENT UNIVERSITY | MSc

Jan 2025 - Jan 2026 | Singapore

Master of Science in Computing

Advisor: ***Prof. Thivya Kandappu***

Courses Taken: *Computational Interaction, Deep Learning and Vision, Learning and Planning in Intelligent Systems, Machine Learning, Optimisation and Computing, Mobile and Distributed Systems*

Research Focus: *Human-Computer Interaction and Creativity Support Tools*

Master's research: ***Motor-Mediated Creativity: Bridging Embodied Skill Training and Digital Expression***

Research resulting in a full paper published at ***ACM CHI 2026.***

UNIVERSITY OF MORATUWA | BSc

Feb 2014 – Jan 2018 | Sri Lanka

B.Sc. Eng. (Hons.) in Electronic and Telecommunication Engineering

Class: ***First class***

Cum. GPA: ***3.70/ 4.20***

Research Experience/ Projects

MOTOR-MEDIATED CREATIVITY: BRIDGING EMBODIED SKILL TRAINING AND DIGITAL EXPRESSION (MOTUS) | Singapore Management University.

April 2025 - Jan 2026 | Singapore

- ➔ Investigated how interactive technologies can scaffold embodied skill development and creative expression without automating the creative outcome.
- ➔ Designed and developed Motus, an interactive creativity-support system combining structured motor training, expert-referenced feedback, and embodied ideation prompts.
- ➔ Implemented real-time stylus interaction and stroke-data capture using a Wacom Movink 13 pen display with a pressure-sensitive stylus.
- ➔ Developed the full system: HTML/CSS/JavaScript frontend and Python backend (Wacom SDK), with real-time stroke capture and multi-modal feedback (ratings, corrective text, graphs and radar plots).
- ➔ Conducted formative and controlled experimental studies examining motor learning, creative expression, user experience, and the effects of feedback and ideation prompting.
- ➔ Applied quantitative and qualitative research methods, including interaction data analysis, questionnaires, expert evaluation, and participant feedback.
- ➔ Contributed to the concept of motor-mediated creativity, examining embodied skill as a resource for creative agency and digital expression (paper published at ACM CHI 2026).

Technologies: HTML, CSS, JavaScript, Python (Integration of Wacom SDK)

IMPROVING FAIRNESS AND ACCESSIBILITY OF CROWD WORK | Singapore Management University.

Nov 2022 - July 2023 | Singapore

- ➔ Developed an AR-based indoor navigation and annotation system supporting accessible interaction for visually impaired users.
- ➔ Designed and implemented an indoor localization tool that used 3D point cloud data to map complex indoor environments, supporting visually impaired individuals in navigation.
- ➔ Developed interactive AR prototypes using Unity, C#, Nreal Light, and 3D point-cloud representations.
- ➔ Explored multimodal interaction techniques including speech, text, hand tracking, and spatial pointers.
- ➔ Conducted a user study to evaluate the system and Contributed to interface prototyping, interaction design, and user-centred development of accessible interactive systems.

Technologies: Unity, C# (Nreal Light AR headset), 3D point cloud processing integration, Figma

GESTURE ENHANCED COMPREHENSION OF AMBIGUOUS HUMAN-TO-ROBOT INSTRUCTIONS | Singapore University of Technology and Design, Agency for Science, Technology and Research (A*STAR), Singapore Management University.

July 2019 - May 2020 | Singapore

- ➔ Designed an HTC Vive-based multimodal interface integrating pointing gestures and natural language to resolve ambiguous instructions in collaborative human-robot tasks.
- ➔ Assisted in experimental studies evaluating gesture-enhanced instruction comprehension and human-robot interaction (paper published at ACM ICMI 2020).

Technologies: Unity (OpenXR), Python, Figma

COMPLEMENTING SPEECH INTERACTION DESIGN WITH TOUCH FOR MULTI-ROBOT SYSTEMS | Singapore University of Technology and Design.

Feb 2018 - July 2019 | Singapore

- ➔ Investigated multi-modal interaction techniques for improving communication between humans and robotic systems.
- ➔ Developed interaction prototypes integrating speech, gesture, touch, handwriting, and other input modalities.
- ➔ Co-designed the interaction interface with the supervising professor and built the multi-modal interfaces (speech, handwriting, gesture, touch, finger mouse, mouthpiece) to assist humans in controlling multiple robots under varying workloads.
- ➔ Conducted research on resolving ambiguous human-to-robot instructions through gesture-enhanced interaction.
- ➔ Worked with Unity, Python, Arduino, Android, and interactive sensing technologies.
- ➔ Ran a user study with 100+ participants, analysed the data, and contributed to writing the paper (paper published at IEEE TENCON 2019).

Technologies: Java (Android Studio), Unity (C#), Python, Arduino, Adobe Photoshop, Figma

Publications

*Bolonghege, P., *Ganganath, G., ***Arachchige, N.,*** Benedict Lincoln, P. and Kandappu, T., 2026, April. Motor-Mediated Creativity: Bridging Embodied Skill Training and Digital Expression. In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (pp. 1-23). **Equal contribution.*
DOI: 10.1145/3772318.3791935

Weerakoon, D., Subbaraju, V., **Karumpulli, N.,** Tran, T., Xu, Q., Tan, U.X., Lim, J.H. and Misra, A., 2020, October. Gesture Enhanced Comprehension of Ambiguous Human-to-Robot Instructions. In Proceedings of the 2020 International Conference on Multi-modal Interaction (pp. 251-259).
DOI: 10.1145/3382507.3418863

Johari, K., **Karumpulli, N.** and Tan, U.X., 2019, October. Complementing Speech Interaction Design with Touch for Multi-Robot Systems. In TENCON 2019-2019 IEEE Region 10 Conference (TENCON) (pp. 1400-1405). IEEE.
DOI: 10.1109/TENCON.2019.8929506

Academic and Research Employment

FACULTY OF ENGINEERING, UNIVERSITY OF RUHUNA | Lecturer (Probationary)
Department of Electrical and Information Engineering
Aug 2026 – Present | Sri Lanka

- ➔ Contribute to undergraduate teaching, curriculum development, student project evaluation, and academic activities within Computer Engineering.
- ➔ Develop practical learning activities in computing, machine learning, and emerging technologies.
- ➔ Supervise and evaluate undergraduate student projects and technical work.
- ➔ Contribute to departmental academic and research activities, with research interests in HCI, Human–AI Interaction, XR, and emerging interactive technologies.

INFORMATICS INSTITUTE OF TECHNOLOGY | Part-Time Lecturer
Jan 2026 – Present | Sri Lanka

- ➔ Deliver lectures for the Professional Certificate in Artificial Intelligence (3-month) program offered by the Professional Development Unit.
- ➔ Teach AI fundamentals including machine learning concepts, data analysis, and real-world AI applications.
- ➔ Develop course materials, assignments, and hands-on practical exercises for students.
- ➔ Design and evaluate assessments and final examinations.

SINGAPORE MANAGEMENT UNIVERSITY | Research Engineer
Nov 2022 – July 2023 | Singapore

- ➔ Conducted HCI research on accessible and multi-modal interactive systems.
- ➔ Designed and developed AR prototypes for indoor navigation and spatial annotation.
- ➔ Implemented multi-modal interaction techniques using speech, text, hand tracking, and spatial interaction.
- ➔ Worked with Unity, C#, Nreal Light, 3D point-cloud data, and Figma.

FACULTY OF ENGINEERING, UNIVERSITY OF RUHUNA | Temporary Instructor
Feb 2022 – July 2022 | Sri Lanka

- ➔ Supported undergraduate teaching and laboratory activities.
- ➔ Assisted students with practical and technical coursework.
- ➔ Contributed to assessment and academic activities within the Faculty of Engineering.

FACULTY OF TECHNOLOGY, UNIVERSITY OF RUHUNA | Temporary Demonstrator
Dec 2021 – Jan 2022 | Sri Lanka

- ➔ Supported undergraduate practical and laboratory teaching activities.
- ➔ Assisted students during laboratory sessions and academic exercises.

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN | Research Assistant
Feb 2018 – April 2021 | Singapore

- ➔ Conducted research on multi-modal Human–Computer and Human–Robot Interaction.
- ➔ Designed and implemented interactive systems combining speech, gesture, touch, handwriting, and other interaction modalities.
- ➔ Developed experimental prototypes using Unity, Python, Android, and Arduino.
- ➔ Contributed to experimental studies, research publications, and interdisciplinary research collaborations.

Awards

• World Prize Winner (Joint - 1st Place) - Chartered Institute of Management Accountants (CIMA) Performance Operations (P1) – November 2013 Examination — **2013/2014**

• Dean's List, Semester 3, 7 and 8, University of Moratuwa — **2015/2017**

• Best Performance at the General Certificate of Education, Advanced Level examination in Physical Science stream (University Entrance) at Southlands College — **2012**

Technical and Research Skills

• **Research Methods:** Controlled user experiments, formative studies, semi-structured interviews, questionnaire design, usability evaluation, interaction logging, quantitative and qualitative analysis, experimental HCI, research-through-design.

• **Human–Computer Interaction:** Human–AI Interaction, creativity support tools, embodied interaction, multimodal interaction, accessible interaction, user-centred design.

• **XR and Interactive Systems:** Unity, C#, AR/VR, OpenXR, Nreal Light, spatial interaction, 3D point-cloud processing.

• **AI and Data:** Python, machine learning, data analysis, generative AI.

• **Multi-modal Interaction:** Speech, text, handwriting, gesture, touch, finger mouse, mouthpiece, pointers, stylus interaction, hand tracking.

• **Web and Research Prototyping:** JavaScript, HTML/CSS, Python-based backends.

• **Design and Prototyping:** Figma, Adobe Photoshop.

• **Additional Technologies:** Java, Android Studio, Arduino.

Professional Qualifications

CHARTERED INSTITUTE OF MANAGEMENT ACCOUNTANTS (CIMA) UNITED KINGDOM — Sep 2012 – May 2015 | Sri Lanka

Passed finalist - Advanced diploma in Management Accounting

References

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Provost's Chair Professor (Teaching),
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